

Understanding and Identifying Design Opportunities for Facilitating Humorous Interactions in Multilingual Multicultural Contexts

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ABSTRACT

Sharing humor within a team can have various benefits. In multilingual teams, however, it is often challenging for nonnative-speaking members (NNS) of the team to fully understand the humor shared among native speakers (NS). In an interview study with 28 NNS, we identified common sources of confusion in humorous interactions. Our study also revealed that the (mis)alignment of people's assumptions of others' knowledge is key to failed humor and its consequences. NNS felt that NS often assumed everyone shared the knowledge required to understand their humor; thus, cultural references unfamiliar to NNS were a dominant source of confusion. On the other hand, NNS often felt ignorant for not knowing the references required to understand NS humor, as they assumed all the references unfamiliar to them must be generic knowledge known by all NS. We propose design ideas for better alignment of common ground assumptions between NS and NNS to facilitate humor-related communication and alleviate possible negative consequence of failed humor.

CCS CONCEPTS

• **Human-centered computing**; • **Collaborative and social computing**; • **Empirical studies in collaborative and social computing**;

KEYWORDS

Multilingual Teams, Humor, Interviews, Common Ground Assumption

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1 INTRODUCTION

Sharing a laugh within a team can have various benefits at the individual, interpersonal and group level. For instance, both verbal and visual humor increase creativity [17] and laughing at the same joke makes individuals like each other more [3]. For groups, shared humor promotes positive group affect through emotional contagion [1] and increases group cohesiveness through the function of humor to negotiate social group boundaries (distinguishing “them” from “us”) [12].

Nevertheless, sharing humor can be challenging for multicultural and multilingual teams. Unlike other forms of informal communication such as small talk, humor “functions as an honest signal of the fact of common knowledge, attitudes and preferences” [6] (p.262), a reliable signal that the humor initiator uses to obtain information about the humor recipients with respect to their possession of shared common knowledge in order to ultimately identify cognitively similar individuals for social affiliation and assortment [6]. Due to lack of common knowledge and language fluency issues, non-native speakers of a common language (NNS) may have difficulties in understanding and appreciating the humor shared among native speakers of that language (NS). Research has suggested that NNS may withdraw from non-essential communication with NS altogether [15][16], and language-based cliques may be formed [11]. Consequently, humor that is intended to promote bonding might end up differentiating those who don't get it from those who do [12]. Especially in teams where the majority are NS, NNS may miss out when the team is perhaps bonding the most around shared humor.

While there is extensive research on task-oriented communication in multilingual multicultural teams (e.g. [4] [6]), humorous interaction in these teams has received little attention. Among the limited existing research investigating intercultural humor, there is still controversy. Some studies have concluded that NNS' lack of shared knowledge need not lead to difficulty in understanding humor (e.g. [12]) and suggest a collaborative and co-constructive view of humor between NS and NNS. Others (e.g. [2]) have identified a lack of lexical and cultural knowledge as two key reasons that account for intercultural failed humor. It is worth noting that both lines of research have primarily used linguistic and discourse analysis on a sample of a few conversations between a few individuals. The different settings where conversations took place (pedagogical versus work) and the different social relationships of the conversation participants may account for the inconsistency in their findings. Further, by focusing on what language behavior was performed, previous research has overlooked what was hidden behind the corpus – what was not performed (e.g., no requests for

clarification) or what was faked (e.g. laughter) – which are important for understanding how NNS experience unshared humor with NS, and the potential emotional and interpersonal consequences of this unshared humor.

Our study makes an original contribution to understanding humorous interaction in multilingual and multicultural context by identifying common sources of confusion perceived by NNS. Additionally, we found that misaligned assumptions about others' knowledge play a key role in both the cause and the consequence of such confusions. NS assumptions about NNS knowledge may have caused them to use references unfamiliar to NNS in their humor; and NNS assumptions about NS knowledge results in their negative emotions and withdrawal from humorous interactions with NS. We discuss how these findings shed light on designing tools that support accurate assumption of others' knowledge.

2 METHOD

We conducted in-depth interviews with 28 nonnative speakers (10 men; 18 women) studying or working at a research university in the US. They were recruited through an internal online participant recruitment system and were given either extra credits or \$5 USD for their time. The participants were from China (15), South Korea (8), Vietnam (3) and Japan (2), and were studying or working in various fields such as computer science, communication, economics, math, etc. Nine were undergraduates, 10 were master's students, 7 were doctoral students, one was a postdoc and one was visiting scholar. They had lived in an English-speaking country for 3 months to 5 years. Their ages ranged from 19 to 41 years old.

Interviews began by asking participants whether there was any specific experience with intercultural humor that came to mind when they made the decision to participate in the study. Then, we asked them to briefly describe any team, research lab, or organization they are part of, have worked or interned at, provide scenarios where humor happened in different settings within these teams (e.g. at a lab meeting, doing work individually, collaborating, casual talk), and provide an account of the content of the humor and what conversation topic led to it. We sought to explore how humor-related nonunderstanding differed from non-humor-related nonunderstanding, and how not understanding humor in English-speaking multicultural contexts different from not understanding humor in their native language and culture.

All the interviews were conducted in English, audio-recorded and transcribed with relevant prosodic information and nonverbals (e.g. laughter, hesitation) noted but speech disfluencies (e.g. false starts) removed from the excerpts for clarity and ease of reading. Each interview lasted 1-2 hours.

We took a grounded theory approach [9] in data analysis. At the open coding stage, the first author exhausted the data and developed an initial set of codes. We took actions to explore boundaries of the codes by paying attention to and actively looking for discrepant data [13]. Through iterative coding and interviewing, initial codes were emerged, broken down, or modified by identification of alternative interpretations and cases that didn't fit. A total of 159 codes and 826 quotes were finalized during the open coding stage. Both authors then further iterated between grouping similar codes, examining

quotes in their context, and uncovering the connections among the constructs and piecing out a framework.

3 FINDINGS

In this section, we present our results on perceived sources of confusion, NNS perceptions of NS assumptions about their knowledge, and NNS assumptions about NS knowledge.

3.1 Perceived sources of confusion

We categorized the perceived sources of confusion mentioned by our participants into language-related, culture-related and both language- and culture-related. We identified two types of language-related sources of confusion: fast speech rate and language use. We also identified four subcategories of culture-related confusions based on how well-known cultural references were perceived to be and how long they were perceived to have existed (further discussed in Section 3.3). Lastly, we specified puns that involve cultural references as one type of confusion related to both language and culture. We present the number of participants mentioning each type of confusion and sample quotes in Table 1. The source of the quotes is indicated by participant number, nationality (China = CN, Japan = JP, South Korea = KR, and Vietnam = VN) and gender (F=female, M=male).

3.2 NNS perceptions of NS assumptions of their knowledge and common ground

One common theme that ran through almost all the interviews with NNS was their perception of NS overestimation of NNS knowledge when making jokes. As is evident from Table 1, participants reported cultural references as the most dominant source of confusion in their humorous interactions with NS. A majority of NNS participants reported that NS often joked around assuming everyone shared common ground with respect to the reference they made, when in fact it was not true. As the following participant noted:

I think they [NS] are assuming that listeners have the same background as the speaker. But the problem is sometimes they don't. So for people like me, maybe their humor may not be that effective as they might imagined. But it's not like there's discrimination or something but they're just naively assuming that we are at the same level as him or her. {P24 KR M}

For NNS, it is therefore much easier to share humor with other international people even though they don't speak the same native language or share much in common. However, the fact that other internationals are aware of their limited common ground makes the communication effective.

The thing about NS is that they don't assume you're from a different country, and they assume people they are talking to have the same American context. For example, they like to refer to one of the most famous American TV shows that they assume everybody in their age would know. But if I talk to international people, if she's from Spain, she understands that even

Table 1: Perceived sources of confusion in humorous interactions with example quotes.

Perceived Source of Confusion	N. of Participants	Examples
Language-related		
Fast speech rate	13	In the case of humor, the person speaks really fast, faster than I could process the meanings, plus there's often multiple meanings in humor. {P16 JP M} The appeal of telling something really interesting especially jokes is that you need to put some twist that is supposed to be surprising. So people will try to speak a bit faster, otherwise it's predictable it won't be funny. {P28 CN F}
Words or phrases with unfamiliar connotations and background, idiomatic language, slang, puns	10	Puns are usually difficult because the ways puns are made requires second understanding of the word, not the literal meaning. And I only get the literal meaning but don't get the connotation of it. {P3 KR F} They talked about "darty", which is like a day party. I thought it was something dirty because of the pronunciation. {P6 CN F} He said I yeeted the ball from the halfway line or something. I think the funny was "yeet". But I didn't get it because the way it's used is a verb like throw. I literally thought it means throw and I thought what's so funny about it. But it's the sound you make when you're excited when making a shot. . . , like a [exclamation]. Yes. Also it's not standardized English. It's from Vine or something. {P19 VN M}
Culture-related		
Generic cultural references, facts about the country and states (Historical figures, holiday traditions, characteristics or stereotypes of states)	16	Someone made a joke about culture differences between American and Canadian. I think it's about Thanksgiving, how they celebrate what they eat for celebration. I don't know much about Thanksgiving tradition, so I didn't really get it. {P13 KR M} People will casually catch up by making jokes about the state, for example, the weather or the general characteristics of people there, like east coast west coast or south north. That is usually a point when I can't get the humor because I definitely know very little about each state in the US. {P18 CN F}
Shared collective memory at certain time in the past (e.g., childhood commercials, brands, policies affected life experience)	11	For humor they like to talk about shows I never watched before, or song or TV drama shows happen in their childhood. It happens a lot. {P10 CN F} They have shared experience that we don't feel relatable. . . they were talking "Oh thanks to Obama we eat healthy lunch" or something. . . it was sarcastic. They blamed Michelle Obama for taking away the cookies and chocolate milk, like ruined their childhood or something. That's something maybe funny only if you experienced it, you went to public school here. {P14 CN M}
Ongoing references and facts (e.g., TV and movie characters, celebrities, news, political figures, games)	28	For instance, Nicki Minaj. They make jokes of NM but I don't know who she is. {P1 CN M} They said some famous line from The Office, "how to (the) turn tables"? I guess everyone has watched so they laughed. {P15 KR F}
Emerging trends, facts and language (new Internet slang, memes, games)	10	(for example,) "sus"? like someone is sus, is suspicious. It's from the game (Among Us) they play. . . I didn't find it that funny maybe (because) I didn't play that game. {P12 VN F}
Related to both language and culture		
Puns involving cultural references	15	Western people like to make pun jokes. They talk a lot about pun jokes especially sports related jokes about sport superstar which I have no knowledge about. {P3 KR F} I do have limitations in understanding all the puns because they might reference to the show or to the 1990s something like a pop singer. {P5 KR F} For puns, most times I get it unless that is politics-related puns, which is beyond the meaning of the words, you have another politics thing to understand. {P25 CN F}

if she speaks about her country I would not understand. So she doesn't speak about it at all and focuses on international subjects we both know, or she tells something about her country and explains it to me. It's a fair situation that she doesn't impose what she

knows as if everyone should know. So it's much easier to understand. {P3 KR F}

Many NNS felt that NS interlocutors did not intentionally make those cultural references in their humor. Some expressed understanding that it is difficult for NS to figure out what an average

or a specific NNS knows by taking their perspective as a NS. One Japanese participant noted:

For native speakers they use the term as usual and they don't expect those terms as difficult phrases. I also didn't know some words to be really hard in Japanese when I was in Japan where there were some international students. I used very easy words for me, but for them it's rare to hear or difficult word for that situation. I think it happens a lot here. Maybe native speakers expect it to be an easy word like kindergarten level English, but for me, for adults or international students, it's very tough. I think it's difficult for native speakers to expect international students cannot understand this word. {P16 JP M}

Some NNS also reported that NS assumptions about NNS knowledge seemed to be all-or-nothing. For instance, quite a few NNS recalled experiences in which NS assumed complete ignorance of NNS based on one or a few instances of their missing out on humor.

If you don't quite get it because it involves some pretty new culture stuff, they (NS) cannot joke with you, right? They will just assume you don't understand any of them. Then they'll treat you like a complete newcomer. When they realized I didn't get it, they tried to explain everything, even basic things to you. But that makes yourself feel really bad. I'm only confused about only one joke. {P17 CN F}

In non-humor-related interactions, it is easy to resolve this misassumption through "grounding", which is the process by which interlocutors coordinate to reach a mutual belief that what has been said has been understood by all parties well enough to move the conversation forward [4]. In case of nonunderstanding, successful grounding typically requires the listener to request for clarification that suggests to the speaker what s/he assumes to be mutually known is not. In humor-related conversations, however, clarification requests are not a common solution as they will "spoil the joke" or "ruin the vibe" and are considered face-threatening to both the speaker and one who requests clarification. Most NNS admitted that they don't know how to deal with unshared humor "professionally".

Work is pretty straightforward. If you don't understand then just ask in a professional manner. I know what to do, versus if it's a joke, I don't know how to handle it in a professionally manner, because if I don't get it and I don't smile, it's offensive. So I feel jokes are pretty stressful compared to if it's work-related where I am able to ask questions and do it professionally. {P20 CN F}

3.3 NNS assumption of NS knowledge

For NNS, part of the face threat of nonunderstanding comes from their assumption that the referenced knowledge in humor must be known to most NS. Most NNS admitted that in their native contexts and language, they have also encountered references they don't know, but are "able to judge if the reference is something everyone [in their native country] is supposed to know" (e.g. P5), or "it is used within certain subculture" (P14) which they are not part of, to

justify their lack of knowledge. In the US and in English, however, they are deprived of such intuitions of what an average American knows that can serve as a benchmark for whether their knowledge is below average. NNS reported they are unable to distinguish generic knowledge from niche knowledge that is only known by a specific population.

I wouldn't know if this is something everyone should know or it's only a few people know. {P21 CN F}

The only benchmark for NNS to judge whether certain reference is well-known among NS so that they also should know is based on how frequent NS refer to it and seem to expect them to know.

I would assume if six different people talk about the same thing with me, they assume me to know it. That's one way of identifying if something is important or essential to know {P18 CN F}

Nevertheless, based on our participants' accounts, cultural references are frequently used in NS humor. NNS tend to assume all the referenced knowledge that eludes them is basic, generic knowledge and is commonly known by all NS. They are compelled to be (at least perceived) equally knowledgeable and smart as NS.

In daily conversations, you think everybody knows about this. You don't want to be perceived as dumb. If everyone around you knows Earth and you ask them "what is Earth?" and that would be a really stupid question. You have this peer pressure to feel included, so you pretend to laugh, to feel like you are intellectually on the same level with everyone else. {P8 KR F}

NNS' possibly inaccurate assumptions that all NS know about certain references may have emotional and interpersonal consequences. For instance, when NNS encounter incomprehensible humor with NS, they report feeling "frustrated" (P24), "stressful" (P3), "hopeless" (P17) and "isolated" (P8) as if "there's an uncrossable barrier that really separates you from the NS" (P8). These negative feelings sometimes can be alleviated when NNS believe that the referenced knowledge is not commonly known among NS. One NNS said:

For new emerging slang, it's new for everybody. There will be Americans also don't understand, then it would feel less embarrassed for me to not get it. But often-times there's no way for me to know if it's new or it's been used for a long time. {P6 CN F}

Evidently, the challenge is to distinguish what knowledge has been around a long time from what is newly emerged, which is just as difficult as distinguishing generic from niche knowledge. These differentiations seemed to matter to our NNS participants, and they used them to judge the importance of knowing certain references.

Based on our data, we identified and visualized four types of knowledge that affect NNS' perceptions of the importance of knowing (see Figure 1). These types of knowledge lie along two correlated dimensions based on NNS assumptions about how well-known the knowledge is among NS and the diffusion of that knowledge over time. The first type of knowledge (A) consists of established language (both formal and informal), generic cultural references and facts about the country. NNS would expect almost every NS in the

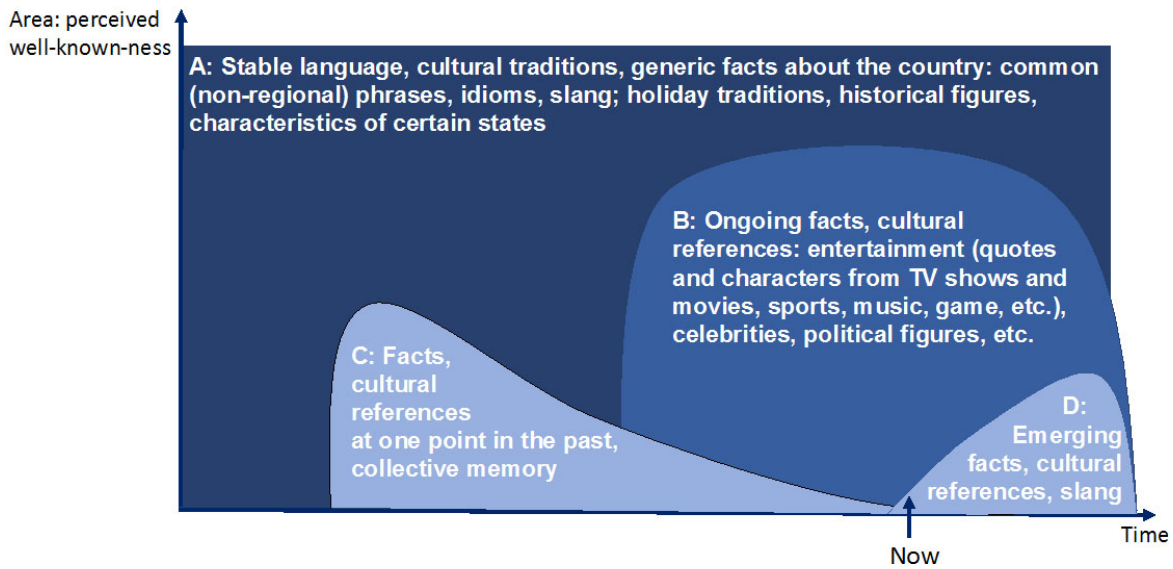


Figure 1: NNS' perceived importance of knowing the referenced knowledge.

U.S. to possess this type of knowledge. The second type of knowledge (B) consists of ongoing facts and cultural references that have arisen in recent decades and are still in effect now and possibly will be in the future. NNS assume many NS in the U.S. know these things, but acknowledge that given the diversity of subcultures, entertainment consumption, online communities and platforms, interest in politics etc., some knowledge and references might be more well-known than others. The third type of knowledge (C) consists of facts or cultural references that were existent at one point in the past but no longer exist or are in effect now, so the number of NS in the U.S. who possess that knowledge will become smaller with time. The fourth type of knowledge (D) consists of emerging facts, language (slang), and cultural references that are trendy and their popularity may grow with time. The larger each area of knowledge is in Figure 1, the more likely NNS will assume that knowledge to be well-known and well-established, the greater their perceived importance of knowing it, and in turn, the more negative the consequence of not knowing that knowledge. The following quote illustrates this:

If it's a very traditional slang or expression, I would attribute to my language ability. I feel lame. Instead of not catching up with new trend. I think there're some really basic things you need to know, for example, expression that people have been using for a long time, or like football, which is something that has been popular in the US for so long, I need to have a basic understanding about. I should know. Culture symbols related to the US, famous restaurants, brands. . . I have this feeling it's very standardized American culture. {P18 CN F}

4 DESIGN IMPLICATIONS

Our findings suggest a potential misalignment of common ground assumptions between NS and NNS. NNS believe that NS's wrong

assumptions about NNS knowledge of certain references in their humor is one cause of their confusion. On the other hand, being unable to distinguish generic knowledge from niche or emergent knowledge, NNS assume all the references NS make in their humor are essential to know. We propose that developing a cultural reference index through crowdsourcing may address this misalignment between NS and NNS for more fluid humor interaction.

For instance, search engines can generate an index of cultural references based on frequency of look-ups by NNS. NNS contributors could choose to provide their record of look-ups, the context where they encountered the reference (e.g. conversation, gaming, memes), the characteristics of the person who brought up the reference (e.g. demographics, subcultures). This index can be used to flag potential lack of understanding or failure of humor in real time, such that the NS who is to make such reference in their humor with NNS could initiate self-correction and explanation in time to avoid further miscommunication; or proactively, to provide context and required knowledge beforehand. The reference index could also be integrated into a conversation agent or a personal agent for NNS that discretely provides information about cultural references on an as-needed basis. This tool might also provide information such as the likelihood of NS knowing the reference, which the NNS could use to decide whether or not to request clarification.

Other NNS could resort to this index to learn about the cultural references they are likely to encounter. NS contributors may also help improve the index to get an accurate estimate of the number of knowers and their demographic characteristics (e.g. teenagers, players of certain game, users of certain platform), the origin and context of the reference, the time it starts to be popular, etc., as these are important factors for NNS to judge if the reference is important to know. With this nuanced information, NNS learning about cultural references can be more targeted. For instance, NNS could choose to thoroughly study generic references and gloss over

those niche references just to heuristically know it's niched the next time they encounter one.

5 LIMITATIONS AND FUTURE WORK

One limitation of this study is that we only have NNS accounts and experiences of unshared humor. Results regarding their perceptions of NS assumptions of NNS knowledge should be treated with caution. In the absence of the NS perspective, it is unclear whether NS actually “assume” NNS knowledge at all when making references in their jokes. However, this assumption seems consistent with previous findings that one's estimate of other's knowledge is biased in the direction of their own, such that one who possesses the knowledge tends to overestimate other's possession of the knowledge [7]. We intend to verify this by involving both NNS and NS perspectives in our future work.

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